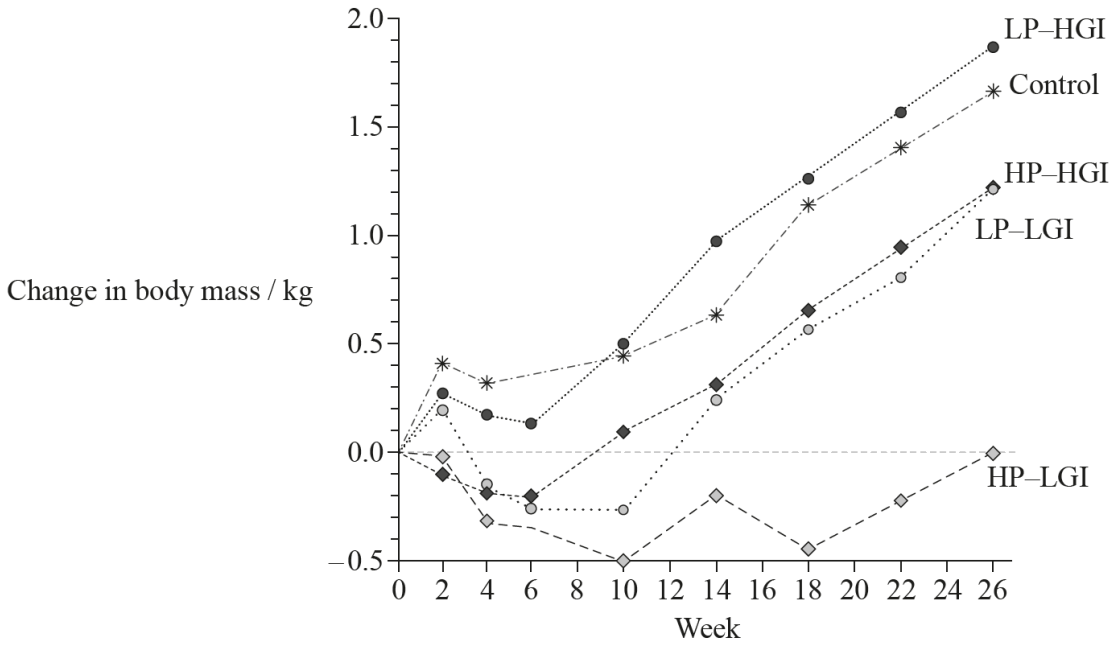
Short questions

|!|BCESB0060000001|!| ★

IBDP Standard level Biology Paper 3 (TZ2) May 2014 Q1

During an investigation, 773 participants who had lost body mass through a low-energy diet were randomly assigned a diet (with no energy restriction) for the next 26 weeks to maintain their mass. These maintenance diets were one of the four combinations of high or low protein with either high or low glycemic index. A control group followed the recommended daily intake. High glycemic index foods enable a quicker release of glucose into the bloodstream than foods with a low glycemic index.



Key:	HP – high protein	HGI – high glycemic index
	LP – low protein	LGI – low glycemic index

[Source: From *The New England Journal of Medicine*, Thomas Meinert Larsen, Stine-Mathilde Dalskov, Marleen van Baak et al., Diets with High or Low Protein Content and Glycemic Index for Weight-Loss Maintenance, vol. 363, pages 2102–2113. Copyright © 2010 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.]

(a) State the change in body mass 10 weeks after the start of the maintenance diet for the low protein and high glycemic index group, giving the units. (1 mark)

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(b) State the trend in body mass during the 26 week period for the maintenance diets with high glycemic index. (1 mark)

(c) Evaluate the effectiveness of a high protein with low glycemic index diet in maintaining mass, after a mass-loss diet. (2 marks)

(d) Explain the possible health consequences of diets rich in carbohydrates. (2 marks)

##

(a) 0.5 kg (1)

(b) Increase / positive trend / towards gain in body mass (1)

(c) The diet is effective at maintaining mass as final mass is equal to initial mass. It is the only group not showing weight re-gain after 26 weeks. (1)
This group shows weight loss / maintenance (with fluctuations) in early weeks, while during last weeks (from week 18) show positive trend. (1)

(d) Any ***two***: (2)

- Excess carbohydrates are converted / stored as fat / lipids. This may lead to weight gain / obesity. (1)
- The diets may increase the risk of diabetes / other related health risk factor. (1)
- Other nutrients may be deficient (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Analysing data	2
(d)	Applying concepts	2

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|!|BCESA0060000002|!| ★★ **STSE**

Plant-based 'meat' is now available in the market. These meat substitutes for vegetarians are commonly made from mixtures of peas, cereals, beans and plant oil. Joe compared the nutrition information of burgers which contain plant-based 'meat' or beef.

	Plant-based 'meat' burger (120 g)	**Beef burger (120 g)**
Total fats	14 g	20 g
Cholesterol	0 mg	90 mg
Sodium	450 mg	150 mg
Total carbohydrates	6 g	0 g
- **Sugars**	0 g	0 g
- **Dietary fibre**	3 g	0 g
Proteins	20 g	21 g

(a) Evaluate the benefits of eating a plant-based 'meat' burger over a beef burger. (3 marks)

(b) Some dietitians are concerned about the sodium content of plant-based 'meat'. What health problem may be caused by an excessive intake of sodium? (1 mark)

##

(a) The plant-based 'meat' burger contains less fat and no cholesterol, which does not increase the risk of having cardiovascular diseases. (1)
The plant-based 'meat' burger contains more dietary fibre, which prevents constipation. (1)
The protein content of a plant-based 'meat' burger is roughly the same as that of a beef burger. Proteins are used by our body for growth and repair. (1)

(b) high blood pressure / kidney disease (1)

(4 marks)

##

	Skills assessed	**Marks**
(a)	Analysing data	3
(b)	Applying concepts	1

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|!|BCESA0060000003|!| ★★ **STSE**

The table below shows the nutrition information of sweet potatoes (per 100 g).

Carbohydrates	20.1 g	
Starch	12.7 g	
Sugars	4.2 g	
Dietary fibre	3.2 g	
Fat	0.1 g	
Protein	1.6 g	
Vitamin A	709 μg	(89% DV*)
Vitamin C	2.4 mg	(3% DV)
Calcium	30 mg	(3% DV)
Iron	0.61 mg	(5% DV)

* DV = daily value

(a) Susan believes that sweet potato is good for health, so she eats only sweet potatoes as breakfast. For her age and body mass, Susan needs 1200 kJ from breakfast. How much sweet potatoes should she eat in order to fulfill this energy requirement? (Hint: One gram of proteins provides 18.2 kJ of energy. One gram of lipids provides 38.9 kJ of energy. One gram of carbohydrates provides 17.1 kJ of energy.) (2 marks)

(b) Deduce which deficiency disease is best treated by eating sweet potatoes. (2 marks)

(c) With reference to the carbohydrate content, describe ***two*** benefits of eating sweet potatoes. (2 marks)

##

(a) Energy content of 100 g of sweet potatoes
= (20.1 × 17.1) + (0.1 × 38.9) + (1.6 × 18.2) = 376.7 kJ (1)
Amount of sweet potatoes needed = (1200 / 376.7) × 100 g = 318.6 g (1)

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(b) Night blindness (1)

because 100 g of sweet potatoes supplies the majority of vitamin A needed by the body per day. (1)

(c) Sweet potatoes contain dietary fibre which can prevent constipation. (1)

Sweet potatoes contain a higher proportion of starch than sugars. As starch has to be digested before absorption, it does not cause immediate increase in the blood glucose level. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	1
	Applying concepts	1
(c)	Analysing data	1
	Applying concepts	1

|!|BCESC0060000004|!|

For each of the food substance listed in column 1, select from column 2 one deficiency disease that is caused by a lack of the food substance. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
Dietary fibre	(a) _____	A. Anaemia
Calcium	(b) _____	B. Constipation
Vitamin A	(c) _____	C. Night blindness
		D. Osteoporosis
		E. Scurvy

##

(a) B (1)

(b) D (1)

(c) C (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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|!|BCESC0060000005|!|

The table below shows the seven types of food substances needed in a balanced diet and their major functions in the body. Complete the table. (5 marks)

Food substance	Major function
(a)	As a main source of energy
(b)	As an energy reserve and insulation
(c)	For growth and repair, making enzymes and antibodies
Vitamins	Help in chemical reactions and are essential for maintaining good health
(d)	Regulate metabolism and as major components of some types of body structures
(e)	Stimulates peristalsis and prevents constipation
Water	As a solvent, medium for chemical reactions and for transport

##

(a) Carbohydrates (1)

(b) Lipids (1)

(c) Proteins (1)

(d) Minerals (1)

(e) Dietary fibre (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0060000006|!| ★ SI

The table below shows the nutritional composition of 100 g of various foodstuffs.

Food	Protein (g)	Carbohydrate (g)	Lipids (g)	Calcium (mg)	Iron (mg)	Vitamin D (IU)
Beef	26	0	12	10.6	3.9	0
Lettuce	1.2	1.5	0	25.1	0.8	0
Pork	25	0	20	10.6	1.1	0
Pig liver	24	1	4.4	7.1	14.1	46

(a) Which food provides the largest amount of energy? (1 mark)

(b) Which ***two*** of the foods are the best in preventing osteoporosis? (2 marks)

(c) Which food is good for people suffering from anaemia? (1 mark)

(d) Outline how you would test for the presence of vitamin C in lettuce. (3 marks)

##

(a) pork (1)

(b) lettuce (1)

pig liver (1)

(c) pig liver (1)

(d) Prepare extracts of lettuce. (1)

Add the lettuce extract drop by drop to DCPIP solution. (1)

If the lettuce extract contains vitamin C, the blue DCPIP solution will be decolourized. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1

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(b)	Analysing data	2
(c)	Analysing data	1
(d)	Practical skills	3

|!|BCESC0060000007|!|

A student performed an experiment to study the action of amylase on starch. He mixed equal volume of 0.1% amylase solution and 1% starch solution in a test tube. Immediately after mixing, he removed three small portions from the reaction mixture and conducted three tests on them. The results are shown in the table below.

Albustix paper test	Benedict's test	Iodine test
+	–	+

Key: + positive result
– negative result

(a) List the possible chemical substance detected by the three tests respectively. (3 marks)

Albustix paper test: _____

Benedict's test: _____

Iodine test: _____

(b) After 60 minutes, he carried out the three tests on the reaction mixture again. Use '+' and '–' to indicate the possible results in the table below. (3 marks)

Albustix paper test	Benedict's test	Iodine test

##

(a) Albustix paper test: proteins (1)
Benedict's test: reducing sugars (1)
Iodine test: starch (1)

(b) Albustix paper test: + (1)
Benedict's test: + (1)
Iodine test: – (1)

(6 marks)

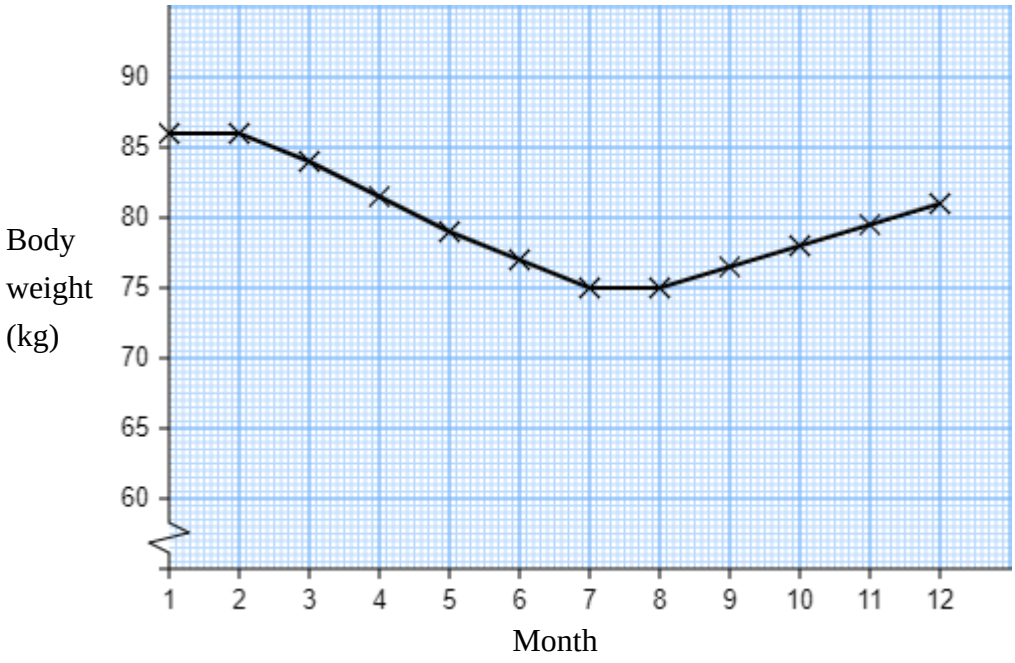
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	3

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|!|BCESB0060000008|!| ★ **STSE**

The graph below shows the change in Jack's weight (in kg) during the course of a year.



Jack consulted a dietician for advice to lose weight. He started to follow the dietician's diet plan on 1st February. He also jogged for 30 minutes every day.

(a) How much weight has he lost by the beginning of June? (1 mark)

_____

(b) Which ***two*** food types in Jack's diet do you think the dietician told him to reduce in quantity? (2 marks)

_____

_____

(c) At the beginning of August, Jack hurt his leg and could not jog any more. He continued with the dietician's diet plan. Explain the change in his body weight after the injury. (2 marks)

_____

_____

##

(a) 9 kg (1)

(b) soft drinks / fried foods / refined carbohydrates / greasy foods / other reasonable answers (Any ***two***, 2)

(c) The energy intake was greater than the energy output. (1)

The excess energy is stored as fat in the body, causing weight gain. (1)

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(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Applying concepts	2

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